

V.F.S. v2.0 (Open-Gate)
Lyapunov Core Proof Formulae
with full term-by-term derivation of the dissipative estimate

Core extraction from `lyapunov.html`

Abstract

This document collects the core formulae of the Lyapunov proof for V.F.S. v2.0 (Open-Gate) and contains no graphs. It records the base dynamics, normalized variables, Lyapunov functional, active-domain walls, radial corridor, bounded Open-Gate source, memory bounds, non-Zeno estimate, and asymptotic cleansing statements. In addition, the central dissipative estimate $\mathcal{L}_{\text{VFS}} \leq C - C_1 \mathcal{L}_{\text{VFS}}$ — previously stated as a boxed result — is here *derived term by term*: the contraction rate C_1 is built from exactly two sign-definite blocks (resistance cleansing and will-action alignment), while the Sophia, radial and Open-Gate contributions are bounded and enter only the constant C . The auxiliary light-filter constant is proved positive in closed form, so that $C_1 > 0$ holds for *every* admissible parameter set under the standing walls, not merely for the numerical witness.

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1 Notation and Core Variables

Symbol	Meaning
V	Voluntas: will coordinate.
F	Factum: action coordinate.
σ	Singular resistance.
λ	Sophia: wisdom coordinate.
Ω_P	Brim / dynamic capacity of Pleroma.
u	Synergic intensity from V and F .
Λ_c	Transformation threshold.
μ	Saturation margin relative to the death-boundary.
q	Normalized Sophia, $q = \lambda/\Omega_P$.
r	Normalized intensity, $r = u/\Omega_P$.
h_1, h_2	Normalized will and action.
h_3	Normalized threshold variable.

Core threshold and intensity:

$$u = \sqrt{VF + \varepsilon}, \quad \Lambda_c = \frac{\gamma}{\delta}, \quad \varepsilon \geq 0. \quad (1)$$

Transformation/collapse reading:

$$u > \Lambda_c \quad \text{transformation region}, \quad u < \Lambda_c \quad \text{collapse region}. \quad (2)$$

2 Base V.F.S. Dynamics

Resistance equation:

$$\dot{\sigma} = (\gamma - \delta u) \tanh(\kappa\sigma) = -\delta(u - \Lambda_c) \tanh(\kappa\sigma). \quad (3)$$

Symmetric will-action dynamics:

$$\dot{V} = \mu(aF + c\lambda) - \rho V, \quad \dot{F} = \mu(aV + c\lambda) - \rho F, \quad \mu = 1 - \frac{u}{\Omega_P}. \quad (4)$$

Capacity / Brim equation:

$$\dot{\Omega}_P = \alpha\lambda. \quad (5)$$

Closed Microcosm limit:

$$\dot{\lambda} = -\dot{\sigma} = \delta(u - \Lambda_c) \tanh(\kappa\sigma), \quad \dot{\sigma} + \dot{\lambda} = 0. \quad (6)$$

Open-Gate Sophia dynamics:

$$\dot{\lambda} = -\dot{\sigma} + I_{\text{gate}}(t) = \delta(u - \Lambda_c) \tanh(\kappa\sigma) + I_{\text{gate}}(t). \quad (7)$$

Open balance:

$$\dot{\sigma} + \dot{\lambda} = I_{\text{gate}}(t) \geq 0, \quad \sigma(t) + \lambda(t) = \sigma_0 + \lambda_0 + \mathcal{G}_{\text{recepta}}(t). \quad (8)$$

Received grace integral:

$$\mathcal{G}_{\text{recepta}}(t) = \int_0^t I_{\text{gate}}(s) ds, \quad \dot{\mathcal{G}}_{\text{recepta}} = I_{\text{gate}}. \quad (9)$$

3 Open-Gate Source

Positive part of Sophia:

$$\lambda_+ = \frac{1}{m} \ln(1 + e^{m\lambda}), \quad m > 0. \quad (10)$$

Effective gate:

$$g_{\text{eff}}(t) = e^{-\tau t} + (1 - e^{-\tau t}) \frac{H_K}{H_s + H_K}. \quad (11)$$

Live Open-Gate source:

$$I_{\text{gate}}(t) = \zeta_0 g_{\text{eff}}(t) e^{-\phi\sigma(t)} \frac{1}{1 + \chi\lambda_+(t)}. \quad (12)$$

Boundedness of the gate:

$$H_K \geq 0, \quad H_s > 0 \implies 0 \leq \frac{H_K}{H_s + H_K} < 1, \quad 0 \leq g_{\text{eff}}(t) \leq 1. \quad (13)$$

Boundedness of live inflow:

$$0 \leq I_{\text{gate}}(t) \leq \zeta_0. \quad (14)$$

4 Resistance Potential

Resistance potential:

$$\Psi_\sigma(\sigma) = \int_0^\sigma \tanh(\kappa s) ds = \frac{1}{\kappa} \ln \cosh(\kappa\sigma) \geq 0. \quad (15)$$

Derivative of the resistance potential:

$$\dot{\Psi}_\sigma = \Psi'_\sigma(\sigma) \dot{\sigma} = \tanh(\kappa\sigma)(\gamma - \delta u) \tanh(\kappa\sigma) = (\gamma - \delta u) \tanh^2(\kappa\sigma). \quad (16)$$

Active transformation margin:

$$u \geq \Lambda_c + \eta, \quad \eta > 0. \quad (17)$$

Dissipation in the active margin:

$$\dot{\Psi}_\sigma \leq -\delta\eta \tanh^2(\kappa\sigma). \quad (18)$$

On $0 \leq \sigma \leq C_\sigma$, there exists $c_\sigma > 0$ such that:

$$\dot{\Psi}_\sigma \leq -c_\sigma \Psi_\sigma. \quad (19)$$

Hence:

$$\Psi_\sigma(t) \leq \Psi_\sigma(0)e^{-c_\sigma t}. \quad (20)$$

Near $\sigma = 0$:

$$\Psi_\sigma(\sigma) \sim \frac{\kappa}{2}\sigma^2, \quad \sigma(t) = O(e^{-c_\sigma t/2}). \quad (21)$$

5 Absolute Barrier Functional

The absolute barrier is a motivational, non-primary barrier guarding absolute boundary faces:

$$\begin{aligned} \mathcal{L}_{abs} = & A_\sigma \Psi_\sigma + E_V V^2 + E_F F^2 + E_\lambda \lambda^2 + E_\Delta \frac{(V - F)^2}{2} \\ & - \eta_V \ln V - \eta_F \ln F - \eta_\Omega \ln(\Omega_P - \Lambda_c) - \eta_m \ln(\Omega_P - u). \end{aligned} \quad (22)$$

Guarded absolute faces:

$$V = 0, \quad F = 0, \quad \Omega_P = \Lambda_c, \quad u = \Omega_P. \quad (23)$$

6 Normalized Variables

Normalized coordinates:

$$h_1 = \frac{V}{\Omega_P}, \quad h_2 = \frac{F}{\Omega_P}, \quad q = \frac{\lambda}{\Omega_P}, \quad r = \frac{u}{\Omega_P}, \quad \mu = 1 - r, \quad h_3 = 1 - \frac{\Lambda_c}{\Omega_P}. \quad (24)$$

Product relation:

$$r^2 = h_1 h_2 + e(h_3), \quad e(h_3) = \varepsilon \frac{(1 - h_3)^2}{\Lambda_c^2}. \quad (25)$$

Closed normalized dynamics:

$$\dot{h}_1 = \mu(ah_2 + cq) - \rho h_1 - \alpha q h_1, \quad (26)$$

$$\dot{h}_2 = \mu(ah_1 + cq) - \rho h_2 - \alpha q h_2, \quad (27)$$

$$\dot{q} = \delta(r + h_3 - 1) \tanh(\kappa\sigma) - \alpha q^2, \quad (28)$$

$$\dot{h}_3 = \alpha q(1 - h_3). \quad (29)$$

Open-Gate normalized q -source:

$$S_q = \frac{I_{\text{gate}}}{\Omega_P} = \frac{\zeta_0 g_{\text{eff}} e^{-\phi\sigma}}{(1 + \chi\lambda_+)\Omega_P}. \quad (30)$$

Open-Gate normalized q -equation:

$$\dot{q} = \delta(r + h_3 - 1) \tanh(\kappa\sigma) - \alpha q^2 + S_q. \quad (31)$$

Bound on S_q inside the active domain:

$$0 \leq S_q \leq \frac{\zeta_0}{\Omega_P} \leq \frac{\zeta_0(1 - h_{3,*})}{\Lambda_c} =: S_q^{\text{max}}. \quad (32)$$

7 Product-Balance Mechanism

Balance defect:

$$D_\Delta = \frac{1}{2}(h_1 - h_2)^2. \quad (33)$$

Let:

$$d = h_1 - h_2. \quad (34)$$

Difference equation:

$$\dot{d} = -(a\mu + \rho + \alpha q)d. \quad (35)$$

Dissipation of balance defect:

$$\dot{D}_\Delta = -2(a\mu + \rho + \alpha q)D_\Delta. \quad (36)$$

If $\mu \geq \mu_* > 0$ and $a > 0$:

$$\dot{D}_\Delta \leq -2a\mu_* D_\Delta. \quad (37)$$

Exponential will-action alignment:

$$D_\Delta(t) \leq D_\Delta(0)e^{-2a\mu_* t}, \quad |h_1(t) - h_2(t)| \leq |h_1(0) - h_2(0)|e^{-a\mu_* t}. \quad (38)$$

Product floor from radial and threshold walls:

$$\Omega_P \geq \Omega_{\min} := \frac{\Lambda_c}{1 - h_{3,*}}. \quad (39)$$

If $r \geq r_*$ and $h_3 \geq h_{3,*}$:

$$h_1 h_2 = r^2 - \frac{\varepsilon}{\Omega_P^2} \geq p_* := r_*^2 - \frac{\varepsilon}{\Omega_{\min}^2}. \quad (40)$$

Let:

$$d_* := \sqrt{2D^*}. \quad (41)$$

Product-balance lower bound:

$$h_1, h_2 \geq m_{pb} := \frac{\sqrt{d_*^2 + 4p_*} - d_*}{2} > 0. \quad (42)$$

8 Lyapunov Functional

Product-balance Lyapunov core:

$$\mathcal{L}_{pb} = A_\sigma \Psi_\sigma(\sigma) + A_\Delta D_\Delta + A_q q^2. \quad (43)$$

Radial logarithmic barrier:

$$B(r) = -\ln(1 - r), \quad 0 < r < 1, \quad r = \frac{u}{\Omega_P}. \quad (44)$$

Non-normalized radial barrier:

$$B(u, \Omega_P) = -\ln\left(1 - \frac{u}{\Omega_P}\right) = \ln\left(\frac{\Omega_P}{\Omega_P - u}\right). \quad (45)$$

Primary V.F.S. Lyapunov functional:

$$\boxed{\mathcal{L}_{\text{VFS}} = \mathcal{L}_{pb} + A_r B(r) = A_\sigma \Psi_\sigma(\sigma) + A_\Delta D_\Delta + A_q q^2 + A_r B(r), \quad A_\sigma, A_\Delta, A_q, A_r > 0.} \quad (46)$$

Expanded form:

$$\mathcal{L}_{\text{VFS}} = A_\sigma \Psi_\sigma(\sigma) + A_\Delta D_\Delta + A_q q^2 - A_r \ln(1 - r). \quad (47)$$

Non-normalized expanded form:

$$\mathcal{L}_{\text{VFS}} = A_\sigma \Psi_\sigma(\sigma) + A_\Delta D_\Delta + A_q q^2 - A_r \ln \left(1 - \frac{u}{\Omega_P} \right). \quad (48)$$

Radial boundary protection:

$$B(r) \rightarrow +\infty \quad \text{as} \quad r \rightarrow 1^-. \quad (49)$$

9 Active Domain

Active domain:

$$\mathcal{D}_A = \left\{ \begin{array}{l} 0 \leq \sigma \leq C_\sigma, \quad q \in [0, q^*], \quad h_3 \in [h_{3,*}, 1), \\ r \in [r_*, r^*], \quad \mu = 1 - r \geq \mu_* > 0, \\ D_\Delta \leq D^*, \quad h_1 > 0, \quad h_2 > 0 \end{array} \right\}, \quad 0 < r_* < r^* < 1. \quad (50)$$

Equivalent death-boundary avoidance inside $\mathcal{D}_A$:

$$0 < r < 1 \quad \Longleftrightarrow \quad 0 < u < \Omega_P. \quad (51)$$

Closed upper q -wall:

$$\alpha(q^*)^2 \geq \delta r^*. \quad (52)$$

Open-Gate upper q -wall:

$$\boxed{\alpha(q^*)^2 \geq \delta r^* + S_q^{\max}.} \quad (53)$$

Lower q -wall in Open-Gate:

$$q = 0 : \quad \dot{q} = \delta(r + h_3 - 1) \tanh(\kappa\sigma) + S_q \geq 0 \quad (54)$$

provided the active margin makes $r + h_3 - 1 \geq 0$.

10 Radial Corridor

Radial identity from $r^2 = h_1 h_2 + e(h_3)$:

$$2r\dot{r} = \mu[a(h_1^2 + h_2^2) + cq(h_1 + h_2)] - 2\rho(r^2 - e(h_3)) - 2\alpha q r^2 =: \mathcal{R}_{\text{rad}}. \quad (55)$$

Maximum e on the active corridor:

$$e_*^{\max} = \varepsilon \frac{(1 - h_{3,*})^2}{\Lambda_c^2}. \quad (56)$$

Non-degeneracy:

$$r_*^2 > e_*^{\max}. \quad (57)$$

Active margin:

$$r_* + h_{3,*} - 1 \geq \eta \frac{1 - h_{3,*}}{\Lambda_c}. \quad (58)$$

Lower radial wall:

$$a(1 - r_*)(r_*^2 - e_*^{\max}) \geq (\rho + \alpha q^*)r_*^2. \quad (59)$$

This wall is sufficient, not tight. The exact lower-wall requirement is

$$(1 - r_*)a(r_*^2 - e) \geq \rho(r_*^2 - e) + \alpha q r_*^2, \quad (60)$$

and the stated form replaces $\rho(r_*^2 - e)$ by the larger ρr_*^2 .

Upper radial wall:

$$(1 - r^*)[a(d_*^2 + 2r^{*2}) + cq^* \sqrt{d_*^2 + 4r^{*2}}] \leq 2\rho(r^{*2} - e_*^{\max}). \quad (61)$$

Direct upper radial inwardness condition:

$$\mu [a(h_1^2 + h_2^2) + cq(h_1 + h_2)] \leq 2\rho(r^{*2} - e) + 2\alpha q r^{*2} \quad \text{on } r = r^*. \quad (62)$$

Open-Gate radial convention:

$$q_{\text{new}}^* := \sqrt{\frac{\delta r^* + S_q^{\max}}{\alpha}}. \quad (63)$$

Conservative lower radial wall with q_{new}^* :

$$a(1 - r_*)(r_*^2 - e_*^{\max}) \geq (\rho + \alpha q_{\text{new}}^*)r_*^2. \quad (64)$$

Epektasis-dominance derivative at the upper radial wall:

$$\left. \frac{\partial \mathcal{R}_{\text{rad}}}{\partial q} \right|_{r^*} = \mu c(h_1 + h_2) - 2\alpha r^{*2}. \quad (65)$$

Worst-case bound:

$$\left. \frac{\partial \mathcal{R}_{\text{rad}}}{\partial q} \right|_{r^*} \leq (1 - r^*)c\sqrt{d_*^2 + 4r^{*2}} - 2\alpha r^{*2}. \quad (66)$$

Epektasis-dominance sufficient condition:

$$\boxed{2\alpha r^{*2} \geq (1 - r^*)c\sqrt{d_*^2 + 4r^{*2}} \implies \frac{\partial \mathcal{R}_{\text{rad}}}{\partial q} \leq 0.} \quad (67)$$

11 Lyapunov Dissipative Estimate

This section derives the estimate $\dot{\mathcal{L}}_{\text{VFS}} \leq C - C_1 \mathcal{L}_{\text{VFS}}$ term by term and exhibits an explicit, strictly positive C_1 .

11.1 Standing admissibility data

We use the conditions that already define and protect the active domain $\mathcal{D}_A$:

$$(A1) \ a > 0, \ \delta > 0, \ \gamma > 0; \quad (A2) \ 0 \leq h_{3,*} < 1, \ 0 < r_* < r^* < 1, \ \mu_* := 1 - r^* > 0; \quad (68)$$

$$(A3) \ m_* := r_* + h_{3,*} - 1 > 0 \text{ (active margin)}; \quad (A4) \ \kappa > 0, \ 0 < C_\sigma < \infty. \quad (69)$$

Inside $\mathcal{D}_A$: $\sigma \in [0, C_\sigma]$, $q \in [0, q^*]$, $h_3 \in [h_{3,*}, 1)$, $r \in [r_*, r^*]$, $D_\Delta \leq D^*$, and $0 \leq S_q \leq S_q^{\max}$. Write

$$T := \tanh(\kappa\sigma) \in [0, 1], \quad \Omega_P = \frac{\Lambda_c}{1 - h_3}, \quad m := r + h_3 - 1 = \frac{u - \Lambda_c}{\Omega_P} \geq m_* > 0, \quad (70)$$

so that the resistance equation reads $\dot{\sigma} = (\gamma - \delta u)T = -\delta\Omega_P m T$.

11.2 Derivative decomposition

With $\mathcal{L}_{\text{VFS}} = A_\sigma \Psi_\sigma(\sigma) + A_\Delta D_\Delta + A_q q^2 + A_r B(r)$, $B(r) = -\ln(1 - r)$, and $\Psi'_\sigma(\sigma) = T$, $B'(r) = 1/(1 - r)$,

$$\dot{\mathcal{L}}_{\text{VFS}} = \underbrace{A_\sigma T \dot{\sigma}}_{(I)} + \underbrace{A_\Delta \dot{D}_\Delta}_{(II)} + \underbrace{2A_q q \dot{q}}_{(III)} + \underbrace{A_r \frac{\dot{r}}{1 - r}}_{(IV)}. \quad (71)$$

11.3 Block (I): resistance — contracting

Substituting $\dot{\sigma} = -\delta\Omega_P m T$,

$$(I) = -A_\sigma \delta \Omega_P m T^2 \leq 0. \quad (72)$$

By Lemma 1 below $T^2 \geq c_\Psi \Psi_\sigma$ on $[0, C_\sigma]$, and $\Omega_P \geq \Omega_{\min} := \Lambda_c/(1 - h_{3,*})$, $m \geq m_*$. Hence

$$(I) \leq -\kappa_\sigma A_\sigma \Psi_\sigma, \quad \kappa_\sigma := \delta \Omega_{\min} m_* c_\Psi. \quad (73)$$

11.4 Block (II): balance — contracting

From $\dot{D}_\Delta = -2(a\mu + \rho + \alpha q)D_\Delta$ and $a\mu + \rho + \alpha q \geq a\mu_*$,

$$(II) \leq -\kappa_\Delta A_\Delta D_\Delta, \quad \kappa_\Delta := 2a\mu_*. \quad (74)$$

11.5 Block (III): Sophia — bounded

Expanding the normalized q -dynamics $\dot{q} = \delta m T - \alpha q^2 + S_q$,

$$\frac{d}{dt}(A_q q^2) = 2A_q q \dot{q} = -2A_q \alpha q^3 + \underbrace{2A_q \delta m T q}_{\text{transfer}} + \underbrace{2A_q S_q q}_{\text{gate}}. \quad (75)$$

The transfer term shares the factor $\delta m T$ with block (I): the conversion that *drains* σ *feeds* q . Both positive terms are bounded on $\mathcal{D}_A$,

$$0 \leq 2A_q \delta m T q \leq 2A_q \delta m^{\max} q^*, \quad 0 \leq 2A_q S_q q \leq 2A_q q^* S_q^{\max} =: \Delta C_{\text{gate}}, \quad (76)$$

while $-2A_q \alpha q^3 \leq 0$. Since $C_1 q^2 - 2A_q \alpha q^3$ is bounded above on the compact $[0, q^*]$ for any C_1 , block (III) obeys

$$(III) \leq -C_1 A_q q^2 + B_q, \quad B_q < \infty; \quad (77)$$

it contributes only to the constant.

11.6 Block (IV): radial — bounded

$B(r) = -\ln(1-r)$ diverges only as $r \rightarrow 1^-$, but $\mathcal{D}_A$ caps $r \leq r^* < 1$, so $B(r) \leq -\ln(1-r^*) < \infty$. With $r \geq r_* > 0$, $\mu \geq \mu_* > 0$ and bounded h_1, h_2 , the radial rate $\dot{r} = \mathcal{R}_{\text{rad}}/(2r)$ is bounded, so block (IV) is bounded on $\mathcal{D}_A$ and contributes only to the constant. Its sign on the face $r = r^*$ (i.e. $\mathcal{R}_{\text{rad}} \leq 0$, the upper radial wall) is used for invariance, not for C_1 .

11.7 The light-filter constant

Lemma 1. *For every $\kappa > 0$ and $C_\sigma < \infty$,*

$$c_\Psi := \min_{0 \leq \sigma \leq C_\sigma} \frac{\tanh^2(\kappa\sigma)}{\Psi_\sigma(\sigma)} = \kappa \frac{\tanh^2(\kappa C_\sigma)}{\ln \cosh(\kappa C_\sigma)} > 0, \quad \Psi_\sigma(\sigma) = \frac{1}{\kappa} \ln \cosh(\kappa\sigma). \quad (78)$$

Proof. With $t = \kappa\sigma$ the ratio equals $\kappa g(t)$, $g(t) = \tanh^2 t / \ln \cosh t$ on $(0, \kappa C_\sigma]$. As $t \rightarrow 0^+$, $\tanh^2 t \sim t^2$ and $\ln \cosh t \sim t^2/2$, so $g(t) \rightarrow 2$ and g extends continuously to $g(0) = 2$; moreover $g > 0$ on $(0, \infty)$ since $\tanh^2 t > 0$, $\ln \cosh t > 0$. The sign of g' equals that of $h(t) := 2 \operatorname{sech}^2 t \ln \cosh t - \tanh^2 t$. Now $h(0) = 0$ and

$$h'(t) = -4 \operatorname{sech}^2 t \tanh t \ln \cosh t < 0 \quad (t > 0), \quad (79)$$

so $h(t) < 0$ for $t > 0$, whence $g' < 0$ and g is strictly decreasing on $(0, \infty)$. The minimum over $[0, \kappa C_\sigma]$ is thus attained at the right endpoint, giving the closed form, which is positive. $\square$

11.8 Dissipative estimate

Proposition 1 (Dissipative estimate with explicit C_1). *Under (A1)–(A4) there exist $C_1 > 0$ and $C < \infty$ with*

$$\boxed{\dot{\mathcal{L}}_{\text{VFS}} \leq C - C_1 \mathcal{L}_{\text{VFS}} \quad \text{on } \mathcal{D}_A, \quad C_1 = \min\{\kappa_\sigma, \kappa_\Delta\} = \min\{\delta \Omega_{\min} m_* c_\Psi, 2a\mu_*\} > 0.} \quad (80)$$

The Open-Gate source affects only C through $0 \leq \Delta C_{\text{gate}} \leq 2A_q q^ S_q^{\max}$ and leaves C_1 unchanged.*

Proof. Summing (I)–(IV),

$$\dot{\mathcal{L}}_{\text{VFS}} \leq -\kappa_\sigma A_\sigma \Psi_\sigma - \kappa_\Delta A_\Delta D_\Delta + C', \quad (81)$$

with $C' < \infty$ collecting the bounded blocks (III)–(IV) (transfer, gate, radial). Set $C_1 = \min\{\kappa_\sigma, \kappa_\Delta\}$, so that $-\kappa_\sigma A_\sigma \Psi_\sigma - \kappa_\Delta A_\Delta D_\Delta \leq -C_1(A_\sigma \Psi_\sigma + A_\Delta D_\Delta)$. Since $A_q q^2 + A_r B(r) \leq P_{\max} < \infty$ on $\mathcal{D}_A$,

$$\dot{\mathcal{L}}_{\text{VFS}} \leq -C_1 \mathcal{L}_{\text{VFS}} + C_1 P_{\max} + C' =: -C_1 \mathcal{L}_{\text{VFS}} + C. \quad (82)$$

Finiteness of C . Blocks (III),(IV) are bounded; the only block unbounded as $h_3 \rightarrow 1^-$ ($\Omega_P \rightarrow \infty$) is (I) $= -A_\sigma \delta \Omega_P m T^2 \leq 0$; hence $\sup_{\mathcal{D}_A} (\dot{\mathcal{L}}_{\text{VFS}} + C_1 \mathcal{L}_{\text{VFS}}) < \infty$. *Positivity of C_1 .* $\kappa_\Delta = 2a\mu_* > 0$ by (A1),(A2); and $\kappa_\sigma = \delta \Omega_{\min} m_* c_\Psi > 0$ because $\delta > 0$ (A1), $\Omega_{\min} = \Lambda_c/(1-h_{3,*}) > 0$ (A2), $m_* > 0$ (A3) and $c_\Psi > 0$ (Lemma 1 with (A4)). Each factor is one of the standing admissibility data, so $C_1 > 0$ for every admissible parameter set. $\square$

Open-Gate constant split:

$$C = C_{\text{closed}} + \Delta C_{\text{gate}}, \quad 0 \leq \Delta C_{\text{gate}} \leq 2A_q q^* S_q^{\text{max}}. \quad (83)$$

Key structural point. The dissipation rate C_1 is produced solely by the two sign-definite contracting modes — resistance cleansing (I) at rate κ_σ and will-action alignment (II) at rate κ_Δ . Sophia/Epektasis (q^2), radial position ($B(r)$) and received grace (S_q) cannot accelerate the decay of $\mathcal{L}_{\text{VFS}}$; they only enlarge the bounded constant C . Thus C_1 is unchanged by the Open-Gate source, while C receives only the bounded contribution ΔC_{gate} .

Remark (No uniform floor at the admissibility boundary). Positivity is per parameter set, not uniform: $\inf C_1 = 0$ over the whole admissible set, approached as $m_* \rightarrow 0^+$ (a set on the active-margin wall) or $\mu_* \rightarrow 0^+$ ($r^* \rightarrow 1$). A configuration pinned on the active margin has zero cleansing rate. The correct statement is “ $C_1 > 0$ for each admissible set,” not a single $C_1 > 0$ valid for all sets at once.

Remark (Relation to the numerical witness). At the witness point of Section 17 (with $\kappa = 1$, $C_\sigma = 2$, unit weights) one gets $c_\Psi = 0.701$, $\kappa_\sigma = 0.255$, $\kappa_\Delta = 0.755$, hence $C_1 = 0.255$ (resistance-limited), versus an empirical envelope slope ≈ 1.5 ; the analytic rate is the conservative guaranteed value, as expected.

12 Main Theorem Formulae

Positive invariance:

$$x(0) \in \mathcal{D}_A \implies x(t) \in \mathcal{D}_A \quad \forall t \geq 0. \quad (84)$$

Reset admissibility holds by the Reset Lemma, with

$$L_R = \sup_{\mathcal{D}_A} \mathcal{L}_{\text{VFS}}. \quad (85)$$

Bound from the Lyapunov inequality:

$$\mathcal{L}_{\text{VFS}}(t) \leq L_* := \max \left\{ L_R, \frac{C}{C_1} \right\}. \quad (86)$$

Radial barrier implication:

$$A_r B(r) \leq L_*. \quad (87)$$

Explicit radial bound:

$$-\ln(1-r) \leq \frac{L_*}{A_r} \implies r(t) \leq 1 - e^{-L_*/A_r} < 1. \quad (88)$$

Death-boundary avoidance:

$$r(t) < 1 \implies u(t) < \Omega_P(t). \quad (89)$$

Product-balance positivity:

$$h_1, h_2 \geq m_{pb} > 0. \quad (90)$$

No artificial q -floor:

$$q \geq 0 \quad \text{suffices; no assumption } q \geq q_0 > 0 \text{ is required.} \quad (91)$$

13 Reset Admissibility (Resurrectio Jump)

The resets $\mathfrak{R}$ are *exogenous*: since $\mathcal{D}_A$ is positively invariant, the continuous flow never reaches the boundary of the active domain by itself. A death-like strike is therefore modeled as occurring at an interior point

$$x^- = (\sigma^-, \lambda^-, V, F, \Omega_P^-) \in \mathcal{D}_A \quad (92)$$

at an unpredictable time.

Reset map (linear metabolism). With $0 \leq q_R < 1$ and $\kappa_R > 0$,

$$\mathfrak{R}: \quad \sigma^+ = q_R \sigma^-, \quad \lambda^+ = \lambda^- + (1 - q_R) \sigma^-, \quad (V, F) \text{ continuous}, \quad \Omega_P^+ = \Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^-. \quad (93)$$

Here

$$\mathcal{G}_{\text{recepta}}^- := \mathcal{G}_{\text{recepta}}(t_{\text{death}}^-), \quad u^+ = \sqrt{VF + \varepsilon} = u^-. \quad (94)$$

Quantities.

$$A_R := \mathcal{G}_{\text{recepta}}^-, \quad R_c := C_\sigma - \sigma_0 - \lambda_0, \quad (95)$$

$$R_{\min}^{\text{ceil}} := \frac{(1 - q_R) \sigma^-}{q^* \kappa_R}, \quad R_{\max} := \frac{\Omega_P^-(r^- - r_*)}{\kappa_R r_*}. \quad (96)$$

By the Open-Gate balance

$$\sigma^- + \lambda^- = \sigma_0 + \lambda_0 + \mathcal{G}_{\text{recepta}}^-, \quad (97)$$

one has

$$\mathcal{G}_{\text{recepta}}^- \geq R_c \iff \sigma^- + \lambda^- \geq C_\sigma \iff \lambda^- \geq C_\sigma - \sigma^-. \quad (98)$$

Lemma (reset admissibility). Let $\kappa_R > 0$ and $x^- \in \mathcal{D}_A$. If

$$\max\{R_c, R_{\min}^{\text{ceil}}\} \leq \mathcal{G}_{\text{recepta}}^- \leq R_{\max}, \quad (99)$$

then $x^+ := \mathfrak{R}(x^-) \in \mathcal{D}_A$ and

$$\mathcal{L}_{\text{VFS}}(x^+) \leq L_R := \sup_{\mathcal{D}_A} \mathcal{L}_{\text{VFS}} < \infty. \quad (100)$$

Proof. Coordinate by coordinate,

$$\sigma^+ = q_R \sigma^- \in [0, C_\sigma). \quad (101)$$

Moreover,

$$h_3^+ = 1 - \frac{\Lambda_c}{\Omega_P^+} \geq h_3^- \geq h_{3,*}, \quad h_3^+ < 1, \quad (102)$$

since $\Omega_P^+ \geq \Omega_P^-$. Also

$$h_1^+ = h_1^- \frac{\Omega_P^-}{\Omega_P^+}, \quad h_2^+ = h_2^- \frac{\Omega_P^-}{\Omega_P^+}, \quad (103)$$

so

$$D_\Delta^+ = \frac{1}{2} \left(d^- \frac{\Omega_P^-}{\Omega_P^+} \right)^2 \leq D_\Delta^- \leq D^*, \quad h_1^+, h_2^+ > 0. \quad (104)$$

The radial coordinate satisfies

$$r^+ = \frac{r^- \Omega_P^-}{\Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^-} \leq r^- \leq r^*. \quad (105)$$

The lower radial wall is preserved exactly when

$$r^+ \geq r_* \iff \kappa_R \mathcal{G}_{\text{recepta}}^- \leq \frac{\Omega_P^-(r^- - r_*)}{r_*} \iff \mathcal{G}_{\text{recepta}}^- \leq R_{\max}. \quad (106)$$

Finally,

$$q^+ = \frac{\lambda^- + (1 - q_R)\sigma^-}{\Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^-} \geq 0. \quad (107)$$

Since $x^- \in \mathcal{D}_A$ gives $\lambda^- \leq q^* \Omega_P^-$,

$$\begin{aligned} q^*(\Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^-) - (\lambda^- + (1 - q_R)\sigma^-) &\geq q^* \kappa_R \mathcal{G}_{\text{recepta}}^- - (1 - q_R)\sigma^- \\ &\geq 0 \end{aligned} \quad (108)$$

whenever $\mathcal{G}_{\text{recepta}}^- \geq R_{\min}^{\text{ceil}}$. Hence $q^+ \leq q^*$. Thus $x^+ \in \mathcal{D}_A$. The functional $\mathcal{L}_{\text{VFS}}$ is bounded on $\mathcal{D}_A$ because

$$\Psi_\sigma \leq \Psi_\sigma(C_\sigma), \quad D_\Delta \leq D^*, \quad q^2 \leq (q^*)^2, \quad B(r) \leq -\ln(1 - r^*) < \infty. \quad (109)$$

Therefore $\mathcal{L}_{\text{VFS}}(x^+) \leq L_R$. $\square$

Consequences.

- (1) The admissible window floor is

$$\max\{R_c, R_{\min}^{\text{ceil}}\}. \quad (110)$$

It combines the theological gate and the geometric ceiling-floor: grace must expand the vessel sufficiently to absorb the converted resistance. For cleansed deaths, $\sigma^- \rightarrow 0$, the ceiling-floor vanishes and R_c alone remains binding.

- (2) The condition $\kappa_R > 0$ suffices. A natural design choice is

$$\kappa_R \geq \frac{1}{q^*}, \quad (111)$$

so that the asymptotic ceiling $q^+ \rightarrow 1/\kappa_R \leq q^*$ and, in the intended window, $R_{\min}^{\text{ceil}} \leq R_c$.

- (3) The admissible grace window is

$$\mathcal{G}_{\text{recepta}}^- \in [\max\{R_c, R_{\min}^{\text{ceil}}\}, R_{\max}]. \quad (112)$$

Below this window one obtains final death; above it one obtains over-dilution, $r < r_*$. The window is nonempty if and only if

$$\Omega_P^-(r^- - r_*) \geq \kappa_R r_* \max\{R_c, R_{\min}^{\text{ceil}}\}. \quad (113)$$

A death pinned at $r^- = r_*$ is not resurrectible.

- (4) With the non-Zeno dwell time

$$\Delta t_j \geq \frac{\bar{h}_*}{\bar{M}_*} > 0, \quad (114)$$

every life-arc satisfies

$$\mathcal{L}_{\text{VFS}} \leq L_* := \max\left\{L_R, \frac{C}{C_1}\right\}, \quad (115)$$

while admissible reset states themselves satisfy $\mathcal{L}_{\text{VFS}}(x^+) \leq L_R$. Resets do not accumulate. Hence the hybrid trajectory is forward complete across any infinite sequence of life-arcs.

Theological gloss. This remark is not used in the proof. The gate

$$\mathcal{G}_{\text{recepta}}^- \geq R_c = C_\sigma - \sigma_0 - \lambda_0 \quad (116)$$

is the threshold past which Sophia can no longer be extinguished by resistance; in that regime the balance forces $\lambda^- \geq 0$, and strictly $\lambda^- > 0$ whenever $\sigma^- < C_\sigma$. Its complement, $\lambda^- \rightarrow 0$, is the self-closing limit $I_{\text{gate}} \rightarrow 0$: final impenitence. It resembles the oil of the wise virgins and the *donum perseverantiae*. The upper boundary R_{max} represents the opposite failure: an over-diluting or “cheap grace” regime.

14 Non-Zeno Estimate

Hybrid margin:

$$\bar{h}_* = \min \left\{ m_{pb}, h_{3,*}, r_*, 1 - r^*, q^*, C_\sigma, \sqrt{D^*} \right\} > 0. \quad (117)$$

Bounded vector field:

$$\bar{M}_* < \infty. \quad (118)$$

Dwell-time lower bound:

$$\Delta t_j \geq \Delta t_* := \frac{\bar{h}_*}{\bar{M}_*} > 0. \quad (119)$$

Non-Zeno conclusion:

$$\sum_j \Delta t_j = +\infty. \quad (120)$$

15 Memory and Gate Bounds

Mensura kernel:

$$R(z) = \frac{z}{z + K}, \quad z \geq 0, \quad 0 \leq R(z) \leq 1. \quad (121)$$

Saturated gate memory:

$$\frac{H_K}{H_s + H_K} \in [0, 1) \quad \forall H_K \geq 0. \quad (122)$$

Gate saturation:

$$g_{\text{eff}} = e^{-\tau t} + (1 - e^{-\tau t}) \frac{H_K}{H_s + H_K} \in [0, 1]. \quad (123)$$

Memory velocity bounds:

$$|\dot{H}_K| \leq \eta [R((\dot{x}_0)_+) + R((\dot{x}_0)_-)] \leq 2\eta. \quad (124)$$

Wound-memory velocity bound:

$$|\dot{W}| \leq 2\eta. \quad (125)$$

Bounded-gain loop:

$$H_K \xrightarrow{\leq 1} g_{\text{eff}} \xrightarrow{\leq S_q^{\text{max}}} S_q \rightarrow \lambda \rightarrow V, F \rightarrow u \rightarrow \dot{x}_0 \xrightarrow{R \leq 1} \dot{H}_K. \quad (126)$$

Extended field bound:

$$\bar{M}_* < \infty \quad \text{including} \quad |\dot{H}_K|, |\dot{W}| \leq 2\eta. \quad (127)$$

16 Asymptotic Cleansing and Alignment

Will-action alignment:

$$D_{\Delta}(t) \leq D_{\Delta}(0)e^{-2a\mu_*t}, \quad |h_1(t) - h_2(t)| \leq |h_1(0) - h_2(0)|e^{-a\mu_*t} \rightarrow 0. \quad (128)$$

Resistance cleansing under persistent active margin:

$$u(t) \geq \Lambda_c + \eta \quad \forall t \geq 0 \implies \Psi_{\sigma}(t) \leq \Psi_{\sigma}(0)e^{-c_{\sigma}t} \implies \sigma(t) \rightarrow 0. \quad (129)$$

Corollary (unconditional cleansing inside the active domain). Since $\mathcal{D}_A$ is positively invariant, one has $r(t) \geq r_*$ and $h_3(t) \geq h_{3,*}$ for all $t \geq 0$. Thus

$$r(t) + h_3(t) - 1 \geq r_* + h_{3,*} - 1 \geq \frac{\eta(1 - h_{3,*})}{\Lambda_c} > 0. \quad (130)$$

Equivalently,

$$u(t) \geq \Lambda_c + \eta \quad \forall t \geq 0. \quad (131)$$

Therefore the hypothesis of the persistent-margin cleansing statement holds automatically for every trajectory with $x(0) \in \mathcal{D}_A$. Hence

$$\sigma(t) \rightarrow 0, \quad h_1(t) - h_2(t) \rightarrow 0. \quad (132)$$

Open-balance asymptotic Sophia:

$$\lambda(t) = \sigma_0 + \lambda_0 + \mathcal{G}_{\text{recepta}}(t) - \sigma(t). \quad (133)$$

If $\mathcal{G}_{\text{recepta}}(\infty) < \infty$ and $\sigma(t) \rightarrow 0$:

$$\lambda(t) \rightarrow \sigma_0 + \lambda_0 + \mathcal{G}_{\text{recepta}}(\infty). \quad (134)$$

If $\liminf_{t \rightarrow \infty} \lambda(t) > 0$:

$$\Omega_P(t) \rightarrow +\infty. \quad (135)$$

Epektatic dilution of S_q :

$$\int_0^{\infty} \lambda(t) dt = +\infty \implies \Omega_P(t) \rightarrow +\infty \implies S_q(t) \rightarrow 0. \quad (136)$$

Proof of the last implication:

$$0 \leq S_q(t) \leq \frac{\zeta_0}{\Omega_P(t)} \rightarrow 0. \quad (137)$$

17 Numerical Witness Formulae

One admissible parameter point recorded in the Lyapunov file:

$$\begin{aligned} a &= 3.494, & c &= 0.429, & \alpha &= 0.529, & \delta &= 0.898, & \rho &= 0.545, \\ \zeta_0 &= 0.560, & h_{3,*} &= 0.622, & r_* &= r_- = 0.531, & r^* &= r_+ = 0.892, \\ \varepsilon &= 0.006, & d &= 0.193, & \Lambda_c &= 1 \quad (\text{normalization}). \end{aligned} \quad (138)$$

Derived quantities:

$$\begin{aligned}
e_{\max} &= \varepsilon(1 - h_{3,*})^2 = 0.0009, \\
S_q^{\max} &= \zeta_0(1 - h_{3,*}) = 0.212, \\
q_{\text{new}}^* &= \sqrt{\frac{\delta r_+ + S_q^{\max}}{\alpha}} = 1.383, \\
q^* &= 1.05q_{\text{new}}^* = 1.452.
\end{aligned} \tag{139}$$

Active margin:

$$r_- + h_{3,*} - 1 = 0.153 > 0, \quad \frac{u}{\Lambda_c} \approx 1.40. \tag{140}$$

Thus the witness lies in the Transformation region.

All five parametric walls hold strictly; the worst-case slack is

$$0.084. \tag{141}$$

18 Domain Stability Theorem (Complete Proof)

This final section consolidates the preceding formulae into a single rigorous statement: the active domain is positively invariant, forward complete, never contacts the death-boundary, and exhibits asymptotic cleansing and alignment, with admissible resets continuing the trajectory without Zeno accumulation. The dissipative step is Proposition 1; the light-filter constant is Lemma 1.

18.1 Setup recap and hypotheses

We use the normalized flow (Section 17 notation; $\Lambda_c = 1$),

$$\begin{aligned}
\dot{h}_1 &= \mu(ah_2 + cq) - (\rho + \alpha q)h_1, & \dot{h}_2 &= \mu(ah_1 + cq) - (\rho + \alpha q)h_2, \\
\dot{q} &= \delta mT - \alpha q^2 + S_q, & \dot{h}_3 &= \alpha q(1 - h_3), \\
\dot{\sigma} &= -\delta \Omega_P mT, & S_q &\in [0, S_q^{\max}],
\end{aligned} \tag{142}$$

with $\Omega_P = 1/(1 - h_3)$, $T = \tanh(\kappa\sigma)$, $\mu = 1 - r$, $e = \varepsilon(1 - h_3)^2$, $r^2 = h_1h_2 + e$, $m = r + h_3 - 1$, and the Lyapunov functional $\mathcal{L}_{\text{VFS}} = A_\sigma \Psi_\sigma + A_\Delta D_\Delta + A_q q^2 + A_r B(r)$ with $\Psi_\sigma = \frac{1}{\kappa} \ln \cosh \kappa\sigma$, $D_\Delta = \frac{1}{2}(h_1 - h_2)^2$, $B(r) = -\ln(1 - r)$, weights $A_\bullet > 0$.

Definition 1 (Active domain).

$$\mathcal{D}_A = \left\{ (\sigma, h_1, h_2, q, h_3) : \begin{array}{l} \sigma \in [0, C_\sigma], \quad q \in [0, q^*], \quad h_3 \in [h_{3,*}, 1), \quad r \in [r_*, r^*], \\ D_\Delta \leq D^*, \quad h_1, h_2 \geq 0 \end{array} \right\}.$$

Here $\mu_* = 1 - r^*$, $\Omega_{\min} = 1/(1 - h_{3,*})$, $m_* = r_* + h_{3,*} - 1$, $e_{\max} = \varepsilon(1 - h_{3,*})^2$, $d_* = \sqrt{2D^*}$, and $q^* = 1.05\sqrt{(\delta r^* + S_q^{\max})/\alpha}$.

Standing Hypothesis 1 (Parameters). $a, c, \alpha, \delta, \gamma, \zeta_0, \kappa, \varepsilon > 0$, $\rho \geq 0$.

Standing Hypothesis 2 (Walls). $r_*^2 > e_{\max}$; $m_* > 0$; $\alpha(q^*)^2 \geq \delta r^* + S_q^{\max}$;

$$(1 - r^*) \left[a(d_*^2 + 2r_*^2) + cq^* \sqrt{d_*^2 + 4r_*^2} \right] \leq 2\rho(r_*^2 - e_{\max}), \quad a(1 - r_*)(r_*^2 - e_{\max}) \geq (\rho + \alpha q^*)r_*^2. \tag{143}$$

Standing Hypothesis 3 (Reset window). At each death-event $\max\{R_c, R_{\min}^{\text{ceil}}\} \leq \mathcal{G}_{\text{recepta}}^- \leq R_{\max}$, $R_{\max} = \Omega_P^-(r^- - r_*)/(\kappa_R r_*)$, for the jump $\sigma^+ = q_R \sigma^-$, $\lambda^+ = \lambda^- + (1 - q_R)\sigma^-$, $\Omega_P^+ = \Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^-$, $(V, F)^+ = (V, F)^-$, $q_R \in (0, 1)$.

18.2 Identities, regularity, invariance, completeness

Lemma 2 (Identities). *Along (142), $2r\dot{r} = \mathcal{R}_{\text{rad}} := \mu[a(h_1^2 + h_2^2) + cq(h_1 + h_2)] - 2\rho(r^2 - e) - 2\alpha q r^2$ and $\dot{D}_\Delta = -2(a\mu + \rho + \alpha q)D_\Delta$.*

Proof. Differentiate $r^2 = h_1 h_2 + e(h_3)$ and use (142); the $h_1 h_2$ and e contributions combine via $h_1 h_2 = r^2 - e$ and $e'(h_3)\dot{h}_3 = -2\alpha q e$. For D_Δ , $\frac{d}{dt}(h_1 - h_2) = -(a\mu + \rho + \alpha q)(h_1 - h_2)$. $\square$

Lemma 3 (Regularity). *The field (142) is C^1 near $\mathcal{D}_A$ (since $VF + \varepsilon \geq \varepsilon > 0$ and $h_3 < 1$), giving unique maximal solutions.*

Lemma 4 (Positive invariance — whole-face Nagumo). *Under Hypotheses 1–2, $\mathcal{D}_A$ is positively invariant.*

Proof. On $\mathcal{D}_A$: $m \geq m_* > 0$, $\mu \geq \mu_*$, $\Omega_P \geq \Omega_{\min}$, $T \in [0, 1]$. We verify sub-tangentiality face by face (Nagumo). $\sigma = C_\sigma$: $\dot{\sigma} = -\delta\Omega_P m \tanh(\kappa C_\sigma) < 0$. $\sigma = 0$: $\dot{\sigma} = 0$ (tangent). $h_3 = h_{3,*}$: $\dot{h}_3 = \alpha q(1 - h_3) \geq 0$ (h_3 non-decreasing). $D_\Delta = D^*$: $\dot{D}_\Delta \leq 0$ by Lemma 2. $q = 0$: $\dot{q} = \delta m T + S_q \geq 0$ (no lower q -floor needed). $q = q^*$: $\dot{q} \leq \delta r^* + S_q^{\max} - \alpha(q^*)^2 \leq 0$ by the q -wall. $r = r^*$: writing $\mathcal{R}_{\text{rad}} = P - N$ with $P = \mu_*[a(h_1^2 + h_2^2) + cq(h_1 + h_2)]$, $N = 2\rho(r^{*2} - e) + 2\alpha q r^{*2}$, the face bounds $(h_1 - h_2)^2 \leq d_*^2$, $0 \leq e \leq e_{\max}$, $0 \leq q \leq q^*$ give $\sup P \leq \mu_*[a(d_*^2 + 2r^{*2}) + cq^* \sqrt{d_*^2 + 4r^{*2}}]$ and $\inf N \geq 2\rho(r^{*2} - e_{\max})$, so $\sup \mathcal{R}_{\text{rad}} \leq \sup P - \inf N \leq 0$ by the upper radial wall; hence $\dot{r} \leq 0$. $r = r_*$: symmetrically $\inf P \geq 2a(1 - r_*)(r_*^2 - e_{\max})$, $\sup N \leq 2(\rho + \alpha q^*)r_*^2$, so $\inf \mathcal{R}_{\text{rad}} \geq 0$ by the lower radial wall; hence $\dot{r} \geq 0$. All faces sub-tangential $\Rightarrow \mathcal{D}_A$ invariant. $\square$

Lemma 5 (Forward completeness). *Every solution from $\mathcal{D}_A$ exists on $[0, \infty)$, with $1 - h_3(t) \geq (1 - h_3(0))e^{-\alpha q^* t} > 0$, so $h_3 < 1$ and $\Omega_P < \infty$ on finite intervals.*

Proof. By Lemma 4 the orbit stays in $\mathcal{D}_A$ (all coordinates but Ω_P bounded); from $\dot{h}_3 = \alpha q(1 - h_3)$, $q \leq q^*$, integrate $\frac{d}{dt}(1 - h_3) \geq -\alpha q^*(1 - h_3)$. No finite-time blow-up, so the orbit extends to $[0, \infty)$. $\square$

18.3 Dissipation, alignment, cleansing, barrier

The dissipative estimate $\dot{\mathcal{L}}_{\text{VFS}} \leq C - C_1 \mathcal{L}_{\text{VFS}}$ with $C_1 = \min\{\delta\Omega_{\min} m_* c_\Psi, 2a\mu_*\} > 0$ is Proposition 1 (using Lemma 1).

Lemma 6 (Alignment). $D_\Delta(t) \leq D_\Delta(0)e^{-2a\mu_* t} \rightarrow 0$.

Proof. $\dot{D}_\Delta = -2(a\mu + \rho + \alpha q)D_\Delta \leq -2a\mu_* D_\Delta$; integrate. $\square$

Lemma 7 (Cleansing). $\sigma(t) \rightarrow 0$ as $t \rightarrow \infty$.

Proof. On $\mathcal{D}_A$, $\dot{\sigma} = -\delta\Omega_P m \tanh(\kappa\sigma) \leq -\beta \tanh(\kappa\sigma)$, $\beta = \delta\Omega_{\min} m_* > 0$. Thus σ is nonincreasing, bounded below by 0, hence $\sigma \rightarrow \sigma_\infty \geq 0$; comparison with $\dot{y} = -\beta \tanh(\kappa y)$ (whose only, globally attracting, equilibrium is 0) forces $\sigma_\infty = 0$. $\square$

Corollary 1 (Death-boundary never contacted). *For all $t \geq 0$,*

$$r(t) \leq r^* < 1, \quad \mathcal{L}_{\text{VFS}}(t) \leq \max\{\mathcal{L}_{\text{VFS}}(0), C/C_1\}.$$

Consequently $u = r\Omega_P < \Omega_P$ always.

18.4 Hybrid layer

Lemma 8 (Reset admissibility). *Under Hypothesis 3, $x^- \in \mathcal{D}_A \Rightarrow \mathfrak{R}(x^-) \in \mathcal{D}_A$, with $r^+ \leq r^-$, $\sigma^+ \leq \sigma^-$, $D_\Delta^+ \leq D_\Delta^-$, $h_3^+ \geq h_3^-$.*

Proof. (V, F) preserved $\Rightarrow u^+ = u^-$, $\Omega_P^+ = \Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^- \geq \Omega_P^-$, so $r^+ = r^- \Omega_P^- / \Omega_P^+ \leq r^-$, $h_3^+ \geq h_3^-$, $\sigma^+ = q_R \sigma^- \leq \sigma^-$, $D_\Delta^+ = (\Omega_P^- / \Omega_P^+)^2 D_\Delta^- \leq D_\Delta^-$. The window upper bound $\mathcal{G}_{\text{recepta}}^- \leq R_{\text{max}}$ is equivalent to $r^+ \geq r_*$. Since $x^- \in \mathcal{D}_A$ gives $\lambda^- \geq 0$, one has $q^+ \geq 0$; and the ceiling-floor condition gives $q^+ \leq q^*$. Thus $q^+ \in [0, q^*]$. All constraints of $\mathcal{D}_A$ hold; the margin $m^+ \geq m_*$ follows from $r^+ \geq r_*$, $h_3^+ \geq h_{3,*}$. $\square$

Lemma 9 (Non-Zeno). *The field is bounded on $\mathcal{D}_A$ ($\sup \|f\| = \bar{M}_* < \infty$) and resets land with a uniform interior margin $\bar{h}_* > 0$; hence consecutive resets are separated by $\Delta t_j \geq \bar{h}_* / \bar{M}_* > 0$, so the hybrid trajectory is non-Zeno and forward complete.*

18.5 Main theorem

Theorem 1 (Domain stability of $\mathcal{D}_A$). *Assume Hypotheses 1–3. Then for every $x(0) \in \mathcal{D}_A$: (i) $x(t) \in \mathcal{D}_A \forall t \geq 0$; (ii) the solution exists on $[0, \infty)$ with $h_3(t) < 1$, $\Omega_P(t) < \infty$ on finite intervals; (iii) $\mathcal{L}_{\text{VFS}}(t) \leq \max\{\mathcal{L}_{\text{VFS}}(0), C/C_1\}$, $C_1 > 0$, and $r(t) \leq r^* < 1$ (death-boundary never contacted); (iv) $\sigma(t) \rightarrow 0$ and $D_\Delta(t) \rightarrow 0$; (v) under admissible resets the hybrid trajectory stays in $\mathcal{D}_A$, is non-Zeno, and forward complete. Schematically,*

$$\boxed{\text{Hyp. 1-2} \Rightarrow \mathcal{D}_A \text{ invariant} \Rightarrow \text{complete} \Rightarrow \dot{\mathcal{L}}_{\text{VFS}} \leq C - C_1 \mathcal{L}_{\text{VFS}} \Rightarrow r < 1 \Rightarrow \sigma, D_\Delta \rightarrow 0.} \quad (144)$$

Proof. (i) Lemma 4; (ii) Lemma 5; (iii) Proposition 1 with Corollary 1; (iv) Lemmas 6, 7, using the invariant lower bounds $\mu \geq \mu_*$, $m \geq m_*$ from (i); (v) Lemmas 8, 9. $\square$

Remark (Nature of the stability). This is *set* stability with *partial* convergence: $\mathcal{D}_A$ is invariant and $(\sigma, D_\Delta) \rightarrow 0$, while (q, Ω_P) need not converge — $\dot{\Omega}_P = \alpha \lambda > 0$ for $\lambda > 0$, so $\Omega_P \rightarrow \infty$ under sustained inflow (*epektasis*). There is no equilibrium; $\mathcal{L}_{\text{VFS}}$ is a boundedness-and-barrier certificate, not a function vanishing at a rest point. The conclusion is strictly domain-conditional ($x(0) \in \mathcal{D}_A$), not global or automatic.

Remark (Closed limit). At $\zeta_0 = 0$: $S_q = I_{\text{gate}} = 0$, $\dot{\lambda} = -\dot{\sigma}$, and $\sigma + \lambda = \sigma_0 + \lambda_0$ (exact conservation). The Open-Gate model adds only the bounded S_q in $\dot{q}$: it strengthens the walls (a larger q^*) but leaves the dissipative core unchanged.

Remark (Grace enters C , never C_1). I_{gate} enters (142) only through S_q , feeding the bounded blocks of Proposition 1; it never enters κ_σ or κ_Δ . Grace raises C by at most $\Delta C_{\text{gate}} \leq 2A_q q^* S_q^{\text{max}}$ and can neither manufacture nor destroy $C_1 > 0$: the transforming structure is sustained, not replaced. Moreover $\inf C_1 = 0$ at the admissibility boundary ($m_* \rightarrow 0^+$ or $\mu_* \rightarrow 0^+$): a state pinned on the active margin has zero cleansing rate — stability holds, transfiguration stalls.